Monomial f(q) = a*qˆ5 Report

Polynomial

$$f(q) = aq^5$$

Naive

$$g_{\text{naive}}(X) = ax^5$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^5 + 10aq^3\sigma^2 + 15aq\sigma^4$$

$$\text{Var}(g_{\text{naive}}(X)) = 25a^2q^8\sigma^2 + 500a^2q^6\sigma^4 + 2850a^2q^4\sigma^6 + 4500a^2q^2\sigma^8 + 945a^2\sigma^{10}$$

$$\text{MSE}(g_{\text{naive}}) = 25a^2q^8\sigma^2 + 600a^2q^6\sigma^4 + 3150a^2q^4\sigma^6 + 4725a^2q^2\sigma^8 + 945a^2\sigma^{10}$$

$$\text{Bias}(g_{\text{naive}}) = 10aq^3\sigma^2 + 15aq\sigma^4$$

Unbiased

$$g_{\text{unb}}(X) = 15a\sigma^4x - 10a\sigma^2x^3 + ax^5$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^5$$

$$\text{Var}(g_{\text{unb}}(X)) = 25a^2q^8\sigma^2 + 200a^2q^6\sigma^4 + 600a^2q^4\sigma^6 + 600a^2q^2\sigma^8 + 120a^2\sigma^{10}$$

$$\text{MSE}(g_{\text{unb}}) = 25a^2q^8\sigma^2 + 200a^2q^6\sigma^4 + 600a^2q^4\sigma^6 + 600a^2q^2\sigma^8 + 120a^2\sigma^{10}$$

Comparison

$$\text{mean gap} = -10aq^3\sigma^2 - 15aq\sigma^4$$

$$\text{variance gap} = -300a^2q^6\sigma^4 - 2250a^2q^4\sigma^6 - 3900a^2q^2\sigma^8 - 825a^2\sigma^{10}$$

$$\text{MSE gap} = -400a^2q^6\sigma^4 - 2550a^2q^4\sigma^6 - 4125a^2q^2\sigma^8 - 825a^2\sigma^{10}$$

$$\text{Bias}(g_{\text{naive}})^2 = 100a^2q^6\sigma^4 + 300a^2q^4\sigma^6 + 225a^2q^2\sigma^8$$